Chapter 2

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Exercise 1.

Show that if a *finite* group G acts on a Hausdorff space Y such that none of its elements has a fixed point, the action is even and thus it yields a Galois cover $Y \to G \setminus Y$ for connected Y.

Proof.

Pick some $y \in Y$. As the space is Hausdorff, for every $gy \neq y$ there exists disjoint open sets U_1, U_g of y and gy respectively. Then as G is finite, we obtain a finite collection of open sets $\{U_g\}$, which are pairwise disjoint open neighbourhoods of the gy's. Then as the collection of open sets is necessarily finite we can take the intersection $\bigcap g^{-1}U_g$ to obtain an open neighbourhood of y whose G-orbit is pairwise disjoint.

It remains to show that distinct elements of G product distinct open sets: It is sufficient to show that $gy \neq g'y$ for all $g \neq g'$; as the Hausdorff condition would imply that $U_g \neq U_{g'}$. But then if such a g, g' exists, the element $gg'^{-1} \neq 1$ would fix the point y. ■

Exercise 2.

Show that a connected cover $Y \to X$ is Galois with automorphism group G if and only if the fibre product $Y \times_X Y$ is isomorphic to the trivial cover $Y \times G \to Y$ (where G carries the discrete topology).

Proof.
I strongly suspect that the converse of the question as stated is not true unless one further assumes G-equivariance on the isomorphism between $Y \times_X Y$ and $Y \times G$. Hence this is something that will be explicitly assumed throughout this proof.

Recall that $Y \times_X Y$ can be characterized as $\{(y, y') \in Y \times Y : \pi(y) = \pi(y')\}$. So each point of $Y \times_X Y$ is exactly the set $\pi^{-1}(x) \times \pi^{-1}(x)$ where $x = \pi(y)$. For the following proof, we will understand $Y \times_X Y$ to be a Y-cover via $\pi_X(y, y') = y$ (by symmetry we can WLOG take the projection on the first coordinate).

$(\Rightarrow)$: Suppose that $Y \times_X Y$ is isomorphic to $Y \times G$ as Y-covers and G-sets. Then G acts simply transitively on the fibers of $Y \times G$; such that the G-equivariance guarantees simple transitivity on the fibers of Y itself.

We make note of a special case: When G is finite, we need not assume equivariance: First, $\deg_X(Y) = \deg_Y(Y \times_X Y) = \deg_Y(Y \times G) = |G|$. But by Proposition 2.2.2., if we fix points $y_0 \in \pi^{-1}(x_0)$, then distinct elements acting on y_0 must product different elements in the fiber. Hence as there are $|G|$ elements acting on a set of size $\deg_X(Y) = |G|$ the action must be transitive.

This special case will become relevant for Chapter 3 Exercise 3(b).

$(\Leftarrow)$: Suppose that Y is Galois. Then G acts transitively on the fibers, such that $\pi_X^{-1}(y)$ is equal to the set $\{(y, gy)\}_{g \in G}$. Hence we get a well-defined bijection [$\phi : Y \times_X Y \to Y \times G$.

We check that ϕ is continuous: Let $U \times V$ be a basic open of $Y \times G$. Then $\phi^{-1}(U \times V) = \{(y, gy) : y \in U, g \in V\} = \bigcup_{g \in G} U \times_X gU$, which is seen to be open.

We check that ϕ is open: Let $U \times_X U'$ be a basic open set of $Y \times_X Y$. Then $\phi(U \times_X U')$ is seen to be a basic open set in $Y \times G$ as well. Since G carries the discrete topology we only need verify the image be open on the Y-coordinate. But then that is exactly U. ■

Exercise 3.

Let X be a connected and locally simply connected topological space and $x \in X$ a fixed base point. Consider a connected cover $p : Y \to X$ and a base point $y \in p^{-1}(x)$.

(a) Show that there is a natural injective homomorphism $\pi_1(Y, y) \to \pi_1(X, x)$.

Proof.
Let $f : [0, 1] \to Y$ be a loop at y. Then by direct composition with the covering map we obtain a loop $p \circ f$ at $p(y) = x$. That this defines a group homomorphism is a routine verification.

Now suppose that $f \in \pi_1(Y, y)$ such that $p \circ f$ is the constant path $\{x\}$. Then the image of f can only land inside $\pi^{-1}(x)$, which is a discrete space. But then the continuous image of a connected set is still connected, so f is necessarily constant; i.e. the homomorphism is injective. ■

(b) Viewing $\pi_1(X, x)^{\mathrm{op}}$ as a subgroup of $\mathrm{Aut}\big(\tilde{X}_x \mid X\big)$ via the isomorphism $\mathrm{Aut}\big(\tilde{X}_x \mid X\big) \cong \pi_1(X, x)^{\mathrm{op}}$, establish an isomorphism $\pi_1(Y, y)^{\mathrm{op}} \setminus \tilde{X}_x \cong Y$.

Proof.
Recall (Proposition 2.4.6) that the isomorphism $\pi_1(X, x) \cong \mathrm{Aut}\big(\tilde{X}_x \mid X\big)^{\mathrm{op}}$ is given by $\alpha \mapsto \phi_\alpha$, where $\phi_\alpha(f) := f \circ \alpha$. This also gives an isomorphism $\pi_1(Y, y) \cong \mathrm{Aut}\big(\tilde{X}_x \mid Y\big)^{\mathrm{op}}$. But then as the embedding $\pi_1(Y, y) \hookrightarrow \pi_1(X, x)$ is defined via composition with the covering map, it necessarily agrees with the isomorphism as the group homomorphism $\mathrm{Aut}\big(\tilde{X}_x \mid Y\big) \to \mathrm{Aut}\big(\tilde{X}_x \mid X\big)$ is defined via composition with the covering map. So $\pi_1(Y, y)^{\mathrm{op}} \setminus \tilde{X}_x \cong \mathrm{Aut}\big(\tilde{X}_x \mid Y\big) \setminus \tilde{X}_x \cong Y$. ■

Exercise 4.

Let G be a connected and locally simply connected topological group with unit element e. Let $\pi : \tilde{G}_e \to G$ be a universal cover, and $\tilde{e}$ the universal element in the fiber $\pi^{-1}(e)$.

(a) Equip $\tilde{G}_e$ with a natural structure of a topological group with unit element $\tilde{e}$ for which π becomes a homomorphism of topological groups.

Proof.

Recall (Construction 2.4.1) that the universal cover $\tilde{G}_e$ is constructed as the set of paths starting at e up to (fixed end point) homotopy. Hence, we define a composition law on $\tilde{G}_e$ as follows: given $f, g : [0,1] \to G$, we let $fg(t) := f(t)g(t)$. That this is indeed a well-defined path follows from the continuity of the group operation and connectedness of G. That this defines a group law with identity element $\tilde{e}$, that π is a group homomorphism and that the diagram below commutes is a route verification. It remains to show continuity of the group operations.

$$\begin{tikzcd} \tilde{G}_e \times \tilde{G}_e \arrow[r, dashed, "\tilde{m}"] \arrow[d, "\pi \times \pi"'] & \tilde{G}_e \arrow[d, "\pi"'] \\ G \times G \arrow[r, "m"] & G \end{tikzcd} \qquad \begin{tikzcd} \tilde{U} \times \tilde{V} \arrow[r, mapsto, dashed] \arrow[d, mapsto] & \widetilde{W} \arrow[d, mapsto] \\ U \times V \arrow[r, mapsto] & W \end{tikzcd}$$

Suppose that $\tilde{f}, \tilde{g} \in \tilde{G}_e$ such that $\tilde{f}\tilde{g} = \tilde{h}$. Let $\widetilde{W}$ denote some open simply connected neighbourhood of $\tilde{h}$. It is then sufficient to show that $\left(\tilde{f}, \tilde{g}\right) \in \tilde{G}_e \times \tilde{G}_e$ admits a basic open neighbourhood contained in $\tilde{m}^{-1}\left(\widetilde{W}\right)$. To that end, by the continuity of multiplication in G, there exists $(f, g) \in U \times V \subseteq m^{-1}(W)$. By shrinking the neighbourhoods U, V if necessary we can WLOG assume they are simply connected and a local trivialization of $\pi \times \pi$. Hence by taking an appropriate lift there exists $\left(\tilde{f}, \tilde{g}\right) \in \tilde{U} \times \tilde{V}$.

It is then sufficient to show that $\tilde{m}\left(\tilde{U} \times \tilde{V}\right) \subseteq \widetilde{W}$. To that end, note that points in $\tilde{U} \times \tilde{V}$ are of the form $\left(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g}\right)$ where $\tilde{\alpha}, \tilde{\beta}$ are paths contained in $\tilde{U}, \tilde{V}$ with startpoints at $\tilde{f}(1)$ and $\tilde{g}(1)$ respectively. Then $\tilde{m}\left(\tilde{\alpha} \circ \tilde{f}, \tilde{\beta} \circ \tilde{g}\right) = \tilde{\alpha}\tilde{\beta} \circ \tilde{f}\tilde{g}$. But then $\alpha\beta$ is a path contained in $U \times V$, such that its image is contained in W. By passing through the lift π, we see that the path $\tilde{\alpha}\tilde{\beta}$ must be contained in $\widetilde{W}$, and we are done.

The proof for showing that inversion is a continuous operation is near-identical, and we omit it here for the sake of brevity. ■

(b) Show that a group structure with the above properties is unique.

Proof.

Suppose that m, m' are two group structures on $\tilde{G}$ that satisfy the given requirements. Then the following must be a commuting diagram of topological groups:

$$\begin{tikzcd} \tilde{G} \times \tilde{G} \arrow[r, shift left, "m'"] \arrow[r, shift right, "m"'] \arrow[dr, dashed] & \tilde{G} \arrow[d, "\pi"] \\ & G \end{tikzcd}$$

but then we are immediately done by Proposition 2.2.2., as $\pi(m'(\tilde{e}, \tilde{e})) = \pi(m(\tilde{e}, \tilde{e})) = e$. ■

(c) Construct an exact sequence of topological groups

$$1 \to \pi_1(G, e)^{\mathrm{op}} \to \tilde{G}_e \to G \to 1,$$

the topology on $\pi_1(G, e)$ being discrete.

Proof.

From (a) we have that π and the embedding of $\pi_1(G, e)^{\mathrm{op}}$ into $\tilde{G}_e$ are well-defined group homomorphisms. As a covering map it is necessarily surjective; and it is immediate that its kernel is $\pi_1(G, e)^{\mathrm{op}}$. We verify that the SES agrees with the given topologies: It is sufficient to show that the topology endowed on $\pi_1(G, e)^{\mathrm{op}}$ as a subspace of $\tilde{G}_e$ is discrete. But as then there is a natural identification of the points in $\pi_1(G, e)^{\mathrm{op}}$ with the elements of $\mathrm{Aut}\left(\tilde{G}_e \mid G\right)$ (Proposition 2.4.6); and the action of $\mathrm{Aut}\left(\tilde{G}_e \mid G\right)$ on $\tilde{G}_e$ is even (Proposition 2.2.3), we see that the induced subspace topology on $\pi_1(G, e)$ is exactly the discrete topology. ■

Exercise 5.

Let $p : Y \to X$ be a cover of a connected and locally simply connected topological space X. For a point $x \in X$, we have defined two canonical left actions on the fibre $p^{-1}(x)$: one is the action by $\mathrm{Aut}(Y|X)$ and the other is that by $\pi_1(X,x)^{\mathrm{op}}$. Verify that these two actions commute, i.e. that $\alpha(\phi(y)) = \phi(\alpha y)$ for $y \in p^{-1}(x)$, $\phi \in \mathrm{Aut}(Y|X)$ and $\alpha \in \pi_1(X,x)$. [*Hint:* Use Theorems 2.3.5 and 2.3.7.]

Proof.
By Theorem 2.3.7. there is a natural identification of the action of $\mathrm{Aut}\big(\tilde{X}|X\big)$ on $p^{-1}(x)$ with the action of $\pi_1(X,x)^{\mathrm{op}}$. But then by Theorem 2.2.10. $\mathrm{Aut}(Y|X)$ is naturally identified with the coset space of $\mathrm{Aut}\big(\tilde{X}|X\big)/\ \mathrm{Aut}\big(\tilde{X}|Y\big)$. Hence the two actions can be identified with the same group action, such that they commute. ■

Exercise 6.

Let $\tilde{X}_\Delta \to X$ be the cover of the connected and locally simply connected space X introduced in Remark 2.4.14 (2).

(a) Verify that the $\pi_1(X,x)$-space corresponding to $\tilde{X}_\Delta$ via Theorem 2.3.4 is $\pi_1(X,x)$ acting on itself via inner automorphisms.

Proof.
Recall that $\tilde{X}_\Delta$ is defined as the pullback with respect to the diagonal map $X \to X \times X$ and the cover $\tilde{X} \to X \times X$ (where $\tilde{X}$ denotes the fundamental groupoid of X). In more concrete terms, the fibre of $\tilde{X}_\Delta$ above $x \in X$ are homotopy classes of loops based at x, with the coarsest topology such that the projection is continuous.

As in the construction of the universal cover (2.4.1), we can give an open neighbourhood basis for $\tilde{x}$ as follows: For $x \in X$, let U be some simply connected neighourhood of x. Then the basic open set $\tilde{U}$ is the set of classes of loops formed by composing $\alpha \in \pi_1(X,x) = \pi^{-1}(x)$ *on both sides* with some path ϕ in X from x to $y \in U$ to obtain $\phi \circ \alpha \circ \phi^{-1} \in \pi_1(X,y) = \pi^{-1}(y)$.

Then as the monodromy action is obtained via lifting a path along successive simply connected neighbourhoods, we obtain that the monodromy action is exactly $\phi \bullet \alpha = \phi \circ \alpha \circ \phi^{-1}$; i.e. inner automorphism. ■

(b) Let $f : [0,1] \to X$ be a path with endpoints $x = f(0)$ and $y = f(1)$. The pullback $f^*\tilde{X}_\Delta$ is a trivial cover of $[0,1]$ by simply connectedness of $[0,1]$, whence an isomorphism of fibres $\tilde{X}_{\Delta,x} \xrightarrow{\sim} \tilde{X}_{\Delta,y}$. Verify that under the identifications $\tilde{X}_{\Delta,x} \cong \pi_1(X,x)$ and $\tilde{X}_{\Delta,y} \cong \pi_1(X,y)$ this isomorphism corresponds to the isomorphism of fundamental groups $\pi_1(X,x) \xrightarrow{\sim} \pi_1(X,y)$ induced by $g \mapsto f \bullet g \bullet f^{-1}$.

Proof.

By chasing the pullback diagram, one finds that $f^*\tilde{X}_\Delta$ is the subspace of $\tilde{X}_\Delta \times [0,1]$ consisting of points of the form $(\pi_1(X, f(t)), t)$; with maps being the projection maps onto each coordinate.

Given any arbitrary cover Y of $[0,1]$ and some point $p \in \pi^{-1}(0)$, consider taking the lift of the identity path $\varphi : [0,1] \to [0,1]$ along p (ie the monodromy action). Then the connected component of p is exactly the image of the lifted path $\tilde{\varphi}_p$, and this allows us to identify the isomorphism $Y_0 \cong Y_1$.

Hence, in the specific case of $f^*\tilde{X}_\Delta$, we find that the isomorphism between $\pi_1(X,x)$ and $\pi_1(X,y)$ is explicitly given by $\alpha \mapsto \tilde{\varphi}_\alpha \bullet \alpha$; which we know to be equal to $\tilde{\varphi}_\alpha \circ \alpha \circ \tilde{\varphi}_\alpha^{-1}$ from the result of part (a). It remains to show that $\tilde{\varphi}_\alpha$ is exactly f. To that end, consider the following commutative diagram:

$$\begin{tikzcd}
 & f^*\tilde{X}_\Delta \arrow[r] \arrow[d] & \tilde{X}_\Delta \arrow[d] \\
{[0,1]} \arrow[ur, dashed, "\tilde{\varphi}_\alpha"] \arrow[r, "\varphi_\alpha"] & {[0,1]} \arrow[r, "f"] & X
\end{tikzcd}$$

where the composite map $[0,1] \to \tilde{X}_\Delta$ is exactly the lift of the path f; which from (a) we know to be the conjugated path $t \mapsto f(tx) \circ \alpha \circ f(tx)^{-1}$. But then this lifts along the connected component of $(\alpha, 0)$ to give us that $\tilde{\varphi}_\alpha \bullet (\alpha, 0) = f \circ (\alpha, 0) \circ f^{-1}$. ■

Exercise 7.

(Deligne) Let X be a connected and locally simply connected topological space. A *groupoid cover* over X is a cover $\pi : \Pi \to X \times X$ together with the following additional data. Denote by s and t the maps $\Pi \to X$ obtained by composing π with the two projections $X \times X \to X$, and consider the fibre product $\Pi \times_X \Pi$ with respect to the map s on the left and t on the right. The groupoid structure is given by three morphisms of spaces over $X \times X$: a multiplication map $m : \Pi \times_X \Pi \to \Pi$, a unit map $e : X \to \Pi$ (where X is considered as a space over $X \times X$ by means of the diagonal map), and an inverse map $\iota : \Pi \to \Pi$ which satisfies $t \circ \iota = s$, $s \circ \iota = t$. These are subject to the following conditions:

$$m(m \times \mathrm{id}) = m(\mathrm{id} \times m), \quad m(e \times \mathrm{id}) = m(\mathrm{id} \times e) = \mathrm{id},$$

$$m(\mathrm{id} \times \iota) = e \circ t, \quad m(\iota \times \mathrm{id}) = e \circ s.$$

A morphism of groupoid covers is a morphism of covers of $X \times X$ compatible with the above additional data.

(a) Verify that the space $\tilde{X} \to X \times X$ of Construction 2.4.13 carries the structure of a groupoid cover over X.

Proof.
The multiplication map m is composition of paths $(f, g) \mapsto f \circ g$, which is well-defined as the fibered product is taking with respect to the projection maps s and t. The unit map e is the diagonal embedding that sends a point to the constant loop at that point. The inverse map is the map that sends a path to the reverse path. That these definitions satisfy the conditions for a groupoid cover is a route verification and is omitted here. ■

(b) Fix a base point (x, x) of $X \times X$ coming from a point $x \in X$. Show that the cover $\tilde{X} \to X \times X$ corresponds via Theorem 2.3.4 to the underlying set of $\pi_1(X, x)$ together with the left action of $\pi_1(X, x) \times \pi_1(X, x)$ given by $(\gamma_1, \gamma_2)\gamma = \gamma_1 \gamma \gamma_2^{-1}$.

Proof.
First notice that $\pi_1(X \times X, (x, x)) = \pi_1(X, x) \times \pi_1(X, x)$ as there is a natural bijection $\mathrm{Hom}([0, 1], X \times X) \cong \mathrm{Hom}([0, 1], X) \times \mathrm{Hom}([0, 1], X)$. Next, notice that $\pi^{-1}(x, x)$ is by definition the collection of homotopy classes of paths from x to x, such that it is directly identified with the set $\pi_1(X, x)$. That the induced monodromy action is exactly as stated is another route verification. ■

(c) Show that for every groupoid cover $\Pi \to X \times X$ there is a unique morphism of groupoid covers $\tilde{X} \to \Pi$.

Proof.
Skipped.

Exercise 8.

Let X be a connected and locally path-connected, but not necessarily locally simply connected topological space. Construct a profinite group G such that the category of finite covers of X becomes equivalent to the category of finite sets equipped with a continuous left G-action.

Proof.
Notice that if Y, Z are two finite covers of X, then the (fibered) product $Y \times_X Z$ is a finite cover of X via Y and Z of degree $\deg(Y)\deg(Z)$. Hence the category of finite covers of X form a filtered inverse system of topological spaces. If in particular one assumes that every finite cover is dominated by a finite Galois cover $(*)$, then by the correspondence of Theorem 2.2.10. we will have that the inverse system of finite covers of X is isomorphic to the inverse system of corresponding automorphism groups. In particular, the inverse limit of the system will be the group G as desired, and the proof of the question statement is analogous to the proof found in Corollary 2.3.9.. Hence we will dedicate the remainder of this proof to proving $(*)$:

Fix some base point $x \in X$, and WLOG suppose that Y is a connected cover of degree n such that $p^{-1}(x) = y_1, ..., y_n$. Now notice that the n-fold product Y_X^n is a degree n^n cover of X, such that it admits an embedding $S_n \hookrightarrow \mathrm{Aut}(Y_X^n \mid X)$ via permutation of the coordinates (it is a well-defined deck transformation as coordinate permutations are isomorphisms of $Y^n \supseteq Y_X^n$).

Let $\hat{Y}$ denote the connected component of the n-fold product Y_X^n that contains the point $(y_1, ..., y_n) \in Y \times_X ... \times_X Y$. Note that as the coordinates are distinct, S_n also embeds into the space $\mathrm{Aut}\left(\hat{Y}|X\right)$. As Y is connected by assumption, the monodromy action of $\pi_1(X, x)$ on Y induces a morphism $\phi : \pi_1(X, x) \to S_n \subseteq \mathrm{Aut}\left(\hat{Y}|X\right)$ such that the image $H := \phi(\pi_1(X, x))$ is a transitive subgroup of S_n. But then the degree of $\hat{Y}$ is exactly equal to the size of the $\pi_1(X, x)$ orbit on the fiber (as connected, locally path connected spaces are necessarily path connected), so ϕ is in fact surjective. We conclude that $\hat{Y}$ is Galois. To finish, notice that $\hat{Y}$ is a finite cover of Y via projection onto the first coordinate. ■

In fact we prove something stronger: That the Galois cover $\hat{Y}$ above is terminal among all Galois covers of Y, such that it is indeed deserving of the name "Galois closure":

Suppose that Z is some Galois cover of X such that its projection factors factors through a cover morphism $f : Z \to Y$. Fix some z_0 such that $f(z_0) = y_1$, and choose $g_i \in \mathrm{Aut}(Z|X)$ such that $f(g_i z_0) = y_i$ (take $g_1 = e$). Then we can define a morphism $\Psi : Z \to Y_X^n$ to be given by $z \mapsto (f(g_1 z), ..., f(g_n z))$ such that $\Psi(z_0) = (y_1, ..., y_n) \in \hat{Y}$. By the connectedness of Z we see that $\Psi(Z) \subseteq \hat{Y}$, and by Lemma 2.2.11. we see that Ψ is in fact a covering map.

Exercise 9.

Let S be a set having at least two elements, and let X be a topological space. Define a presheaf $\mathcal{F}_S$ of sets on X by setting $\mathcal{F}_S(U) = S$ for all nonempty open sets $U \subset X$ and $\rho_{UV} = \mathrm{id}_S$ for all open inclusions $V \subset U$. Give a necessary and sufficient condition on X for $\mathcal{F}_S$ to be a sheaf.

Proof.

The author has made the strange choice in defining sheaves that two sections need only be glued together when they agree on a *nonempty* intersection (Definition 2.5.4.). I will be ignoring this convention.

We show that the necessary and sufficient condition is for every pair of (non-empty) open sets to have nonempty intersection.

Necessary: Suppose to the contrary that $U \cap V = \varnothing$ for nonempty open sets U, V. Then as $|S| \geq 2$, pick $s, s' \in S$ distinct. Then the sections $s \in \mathcal{F}_S(U)$ and $s' \in \mathcal{F}_S(V)$ cannot be glued to a section on $U \cup V$, so $\mathcal{F}_S$ cannot be a sheaf.

Sufficient: Suppose we have a collection of sections (s_i, U_i) that agree on intersections. We show that this implies that every section must in fact be the same element of S: Choose two sections s_i, s_j. Then $U_i \cap U_j \neq \varnothing$, so $s_i = s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} = s_j$. But then $U_i \cap U_j \subset U_i$ and $U_i \cap U_j \subset U_j$ so we get that $s_i = s_j$. ■

Exercise 10.

Let X be a topological space. Show that the category of sheaves on X is equivalent to the category of those spaces $p : Y \to X$ over X where the projection p is a local homeomorphism (i.e. each point in Y has an open neighbourhood such that $p|_U$ is a homeomorphism onto its image). [*Hint:* Begin by showing that for a sheaf $\mathcal{F}$ the projection $p_{\mathcal{F}} : X_{\mathcal{F}} \to X$ is a local homeomorphism.]

Proof.

We prove this via the construction of a functor and its inverse. We first show that $\mathcal{F} \mapsto X_{\mathcal{F}}$ is a well-defined functor:

Let $\mathcal{F}$ be some sheaf over X. Let $s = [(U,t)]_x \in \mathcal{F}_x$ denote some arbitrary point of $X_{\mathcal{F}}$. Then we have an associated induced map $i_s : U \to X_{\mathcal{F}}$, which is continuous by definition, which we see to be a homeomorphism between U and $i_s(U)$. Indeed, $i_s \circ p_{\mathcal{F}} = p_{\mathcal{F}} \circ i_s = \mathrm{id}$.

Now suppose that $\Phi : \mathcal{F} \to \mathcal{G}$ is some morphism of sheaves. Then define a map $\phi : X_{\mathcal{F}} \to X_{\mathcal{G}}$ to be given by $[(U,t)]_x \mapsto [(U,\Phi(t))]_x$. That this is indeed well-defined follows from the compatibility definition of a morphism. That this map commutes with the respective projections onto X follows from construction. Finally it is continuous as $\phi^{-1}\big(i_{\Phi(s)}(U)\big) = i_s(U)$.

We next show that $Y \mapsto \mathcal{F}_Y$ is a well-defined functor:

That $\mathcal{F}_Y$ is a well-defined sheaf follows from Proposition 2.5.8.. Now suppose that $\Psi : Y \to Z$ is a morphism of étale spaces over X. Then for U open in X, define $\psi : \mathcal{F}_Y(U) \to \mathcal{F}_Z(U)$ to be given by $s \mapsto \Psi \circ s$. As this map is given by left composition of a function, that is automatically guaranteed to satisfy the compatibility condition on restrictions, so it induces a well-defined morphism of sheaves.

We now show that $\mathcal{F} \mapsto \mathcal{F}_{X_{\mathcal{F}}} \cong \mathrm{id}$:

To $s \in \mathcal{F}(U)$, we associate the section $i_s : U \to X_{\mathcal{F}}$. This induces a well-defined morphism of sheaves, as restriction on sections in $\mathcal{F}(U)$ correspond to restricting the domain of i_s. As $\mathcal{F}$ is assumed to be a sheaf, differing sections must have at least one differing germ; such that this association is injective. Now let $i : U \to X_{\mathcal{F}}$ be a section with $V \subseteq i(U)$ mapped homeomorphically down to $p(V)$. Then $i^{-1}(V) = p(V)$, such that $i|_V$ is a local homeomorphism. So the association is surjective and an isomorphism. That this defines a natural transformation is a route verification.

We finally show that $Y \mapsto X_{\mathcal{F}_Y} \cong \mathrm{id}$:

For $y \in Y$, let V be a neighbourhood of y that is mapped homeomorphically down to U. Then to y we associate the germ $[(U,s)]_{p(y)}$, where $s : U \to V$ is the corresponding section. This association is seen to be injective, as differing y will always either have distinct stalks or germs. It is also seen to be surjective, as $[(U,t)]_x = [(U,t)]_{p(y)}$ always for some $p(y)$, where we shrink U if necessary to obtain the section $t : U \to V \ni p(y)$ as needed. Finally we see that this association defines a natrual transformation in Y, as morphisms $Y \to Z$ induce morphisms $\mathcal{F}_Y \to \mathcal{F}_Z$ via post-composition of sections; and the induced morphism $X_{\mathcal{F}_Y} \to X_{\mathcal{F}_Z}$ can be seen to commute with the resultant diagram. ∎

Exercise 11.

Let $\mathcal{F}$ be a presheaf of sets. Show that there is a natural morphism of presheaves $\rho : \mathcal{F} \to \mathcal{F}_{X_{\mathcal{F}}}$, and moreover every morphism $\mathcal{F} \to \mathcal{G}$ with $\mathcal{G}$ a sheaf factors as a composite $\mathcal{F} \xrightarrow{\rho} \mathcal{F}_{X_{\mathcal{F}}} \to \mathcal{G}$. [*Remark:* For this reason $\mathcal{F}_{X_{\mathcal{F}}}$ is called the *sheaf associated with* $\mathcal{F}$.]

Proof.
We have that any morphism of presheaves $\mathcal{F} \to \mathcal{G}$ induces a morphism $X_{\mathcal{F}} \to X_{\mathcal{G}}$ of spaces over X in a functorial manner (see p.50). In particular, by using the equivalence of categories proved in Exercise 10, this in turn induces a morphism $\mathcal{F}_{X_{\mathcal{F}}} \to \mathcal{F}_{X_{\mathcal{G}}}$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathcal{F} \arrow[r] \arrow[d] & \mathcal{G} \arrow[d] & & \\
\mathcal{F}_{X_{\mathcal{F}}} \arrow[r] & \mathcal{F}_{X_{\mathcal{G}}} & \cong & \mathcal{G}
\end{tikzcd}$$

where $\mathcal{F}_{X_{\mathcal{G}}} \cong \mathcal{G}$ by the equivalence. ■